

AE646 — Individual Contribution Statement & AI Tool Use Declaration

Team PINNacles

Aryamann Srivastava

November 2026

Individual Contribution Statement

Team member: Aryamann Srivastava

- **Problem formulation and literature review:** Identified the 2D Darcy flow benchmark in PDEBench as the project focus; reviewed the FNO paper [1] and PDEBench paper [2].
- **Data pipeline:** Implemented `src/download_data.py` and `src/preprocess.py` for downloading and preprocessing PDEBench data; designed the 8000/900/200 train/val/test split.
- **Model implementation:** Implemented `MLPBaseline`, `SpectralConv2d`, and `FNO2d` in `src/models.py`; verified correctness against the reference FNO codebase.
- **Training:** Implemented `src/train.py` with Adam optimiser, cosine LR annealing, and early stopping; ran all three training runs (FNO original, MLP, FNO improved) on the lab GPU VM.
- **Evaluation:** Implemented `src/evaluate.py`, `src/superres_eval.py`, and `src/benchmark_speed.py`; collected all quantitative results.
- **Component profiling:** Designed and implemented `src/benchmark_components.py` with correct GPU synchronisation to measure per-layer execution time.
- **Figure generation:** Implemented `src/generate_figures.py`; produced all six report figures from stored JSON results.
- **Reports and presentations:** Wrote Stage 1 proposal (`PROPOSAL.tex`), Stage 1 presentation (10 slides), Stage 2 interim report (`INTERIM_REPORT.tex`), and this final report (`FINAL_REPORT.tex`).

This is a solo project; all contributions above are by Aryamann Srivastava.

AI Tool Use Declaration

In accordance with the AE646 course policy on generative AI tools, I declare the following use of AI assistance during this project:

1. **Code debugging:** Claude Code (Anthropic) was used to identify and fix a GPU timing bug in the component profiling script. Specifically, the issue was that `time.perf_counter()` measured dispatch time rather than execution time for asynchronous GPU operations; the

fix (`torch.cuda.synchronize()` before each timer call) was suggested by the AI assistant. All code was reviewed, understood, and accepted by me before being committed.

2. **LaTeX formatting:** Claude Code assisted with LaTeX document structure and table formatting for the proposal and reports. All technical content, numbers, and scientific claims are my own.
3. **Presentation slides:** Claude Code assisted with Python code to programmatically build the Stage 1 PPTX presentation using `python-pptx`. The technical content of every slide was written by me.
4. **No AI-generated results:** All numerical results in this report are from experiments I ran on real hardware. No results were generated or hallucinated by AI tools.

Signature: Aryamann Srivastava

Date: November 2026

References

- [1] Z. Li et al., “Fourier Neural Operator for Parametric Partial Differential Equations,” *ICLR*, 2021.
- [2] M. Takamoto et al., “PDEBench,” *NeurIPS*, 2022.